Practical : 2	Depth-First Search (DFS) Using Python..
Date : 10/08/26	

Aim: -

To implement the Depth-First Search (DFS) algorithm using Python to traverse a graph representing different departments of a bank and understand how DFS explores one branch completely before moving to another branch.

Algorithm:-

1. Start the program.
2. Import the required libraries.
3. Create a graph representing the bank and its departments.
4. Create an empty set to store visited nodes.
5. Define a recursive DFS function.
6. Start the traversal from the Bank node.
7. Mark the current node as visited.
8. Visit each connected department recursively.
9. Continue exploring one branch completely before returning to explore another branch.
10. Repeat until all connected nodes are visited.
11. Display the final traversal and total number of departments visited.
12. Stop the program.

Program:-

```
# ============================================================
# Practical No. 2
# Depth First Search (DFS)
# Real Life Example - Bank Department Visit
# ============================================================

import time
import os
import datetime

# Display Program Title
print("=" * 65)
print(" DEPTH FIRST SEARCH (DFS)")
print("=" * 65)

# Display Date and Time
current = datetime.datetime.now()
print("\nDate :", current.strftime("%d-%m-%Y"))
print("Time :", current.strftime("%H:%M:%S"))

# Display Operating System
print("\nOperating System :", os.name)

# Create Bank Graph
bank = {
    "Bank": ["Account Section", "Loan Section"],
    "Account Section": ["Saving Account", "Current Account"],
    "Saving Account": [],
    "Current Account": [],
    "Loan Section": ["Home Loan", "Car Loan"],
    "Home Loan": [],
    "Car Loan": []
}

# Display Graph
print("\nBank Department Structure")
print("-" * 40)

for node in bank:
    print(node, "->", bank[node])

# Create Empty Visited Set
visited = set()

# DFS Function
def dfs(graph, node):
    if node not in visited:

        # Mark node as visited
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```
    visited.add(node)

    # Display current node
    print("\n-------------------------------------")
    print("Current Department :", node)
    print("Visited Nodes :", visited)

    time.sleep(1)

    # Visit every connected department
    for neighbour in graph[node]:
        print("Moving To :", neighbour)
        time.sleep(1)
        dfs(graph, neighbour)


# Start DFS Traversal
print("\n" + "=" * 65)
print("DFS Traversal Started")
print("=" * 65)

start_node = "Bank"

print("\nStarting Department :", start_node)
time.sleep(2)

dfs(bank, start_node)

# Traversal Completed
print("\n" + "=" * 65)
print("DFS Traversal Completed")
print("=" * 65)

print("\nFinal Traversal")
for department in visited:
    print(department)

print("\nTotal Departments Visited :", len(visited))
print("\nProgram Completed Successfully.")
print("=" * 65)
```

Output:

```
 DEPTH FIRST SEARCH (DFS)
================================================================

Date : 23-08-2026
Time : 22:59:50

Operating System : nt

Bank Department Structure
----------------------------------------
Bank -> ['Account Section', 'Loan Section']
Account Section -> ['Saving Account', 'Current Account']
Saving Account -> []
Current Account -> []
Loan Section -> ['Home Loan', 'Car Loan']
Home Loan -> []
Car Loan -> []

================================================================
DFS Traversal Started
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Starting Department : Bank

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Current Department : Bank
Visited Nodes : {'Bank'}
Moving To : Account Section

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Current Department : Account Section
Visited Nodes : {'Bank', 'Account Section'}
Moving To : Saving Account

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Current Department : Saving Account
Visited Nodes : {'Bank', 'Saving Account', 'Account Section'}
Moving To : Current Account
```

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Current Department : Current Account
Visited Nodes : {'Bank', 'Saving Account', 'Current Account', 'Account Section'}
Moving To : Loan Section

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Current Department : Loan Section
Visited Nodes : {'Current Account', 'Bank', 'Account Section', 'Saving Account', 'Loan Section'}
Moving To : Home Loan

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Current Department : Home Loan
Visited Nodes : {'Current Account', 'Bank', 'Account Section', 'Saving Account', 'Home Loan', 'Loan Section'}
Moving To : Car Loan

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Current Department : Car Loan
Visited Nodes : {'Current Account', 'Bank', 'Account Section', 'Saving Account', 'Home Loan', 'Loan Section', 'Car Loan'}

================================================================
DFS Traversal Completed
================================================================

Final Traversal
Current Account
Bank
Account Section
Saving Account
Home Loan
Loan Section
Car Loan

Total Departments Visited : 7

Program Completed Successfully.
================================================================
```

Conclusion:

Thus, the **Depth-First Search (DFS) algorithm was successfully implemented using Python**. The bank department graph was traversed using recursion, where DFS explored one branch completely before backtracking to visit the remaining branches.